

G Pulse

A synthesis drum machine: eight voices, four character models and a 32-step sequencer.



Overview

G Pulse builds every drum sound from scratch - there are no samples. Eight voices (KICK, SNARE, CLAP, CL HAT, OP HAT, TOM, PERC and CYMBAL) are played by a 32-step sequencer that runs with REAPER's transport, or from MIDI drum notes.

One CHARACTER switch changes how the whole kit is made. ANALOG is classic virtual analogue in the spirit of the 80s drum machines: pitch-swept sines, filtered noise and six-square metal for the hats and cymbal. FM uses two-operator FM with its own index envelope. PHYSICAL strikes modelled membranes, plates, wooden bars and hands with a mallet, and the snare has wires that buzz with the head. DIGITAL uses phase distortion, sample-and-hold noise and bit and sample-rate reduction.

Every row of the sequencer can have its own length, so patterns can drift against each other and line up again (polyrhythm). A bus compressor glues the kit, and the pattern can be sent out as MIDI notes to record.

Reach for it for

- Programming drum parts right inside REAPER, without samples

- Classic analogue-style kits, FM clicks and metal, or lo-fi digital grit
- Polyrhythmic patterns with rows of different lengths
- Playing synthesised drums from a MIDI controller or a MIDI item
- Turning a sequenced pattern into MIDI notes you can edit

Getting started

1. Insert a new track and add G Pulse to it. It is an instrument: it makes its own sound and needs no audio input.
2. Make sure PLAY (in TRANSPORT) is lit, then press play in REAPER. The sequencer follows the project's tempo and position, so it plays in time and restarts where the transport does.
3. Click a voice's name to hear it, pick a pattern slot (1 - 8), and edit the steps in the grid.
4. To play it from MIDI instead, arm the track with a MIDI input and record monitoring on, or put a MIDI item on the track: G Pulse responds to General MIDI drum notes (see below). Switch PLAY off if you want MIDI only.

Controls

The header

ANALOG / FM / PHYSICAL / DIGITAL	The CHARACTER model for the whole kit. Each model has its own recipe for every voice; the levels are balanced so switching keeps the kit roughly even.
Display	The model and the pattern slot (with * when it has unsaved edits). While playing it shows the bar, step and tempo; with PLAY on and the transport stopped it reads WAITING FOR TRANSPORT; with PLAY off, SEQUENCER OFF - MIDI ONLY.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Sequencer

Eight rows of 32 sixteenth-note steps. A step is a rest, a hit or an accent (louder, marked with a white bar).

Steps	Click a step to set or clear it; drag to paint along the row. Shift-click sets an accent.
Row length	Right-click a step to end the row there. Every row can have its own length, from 1 to 32 steps; steps past the end are dimmed and a line in the voice's colour marks the end.
VOICE name	Click it to hear the voice. The name flashes in the voice's colour each time it plays.
M	Mutes the row.
LEN	The row's length in steps. Drag up or down on it, or use the mouse wheel.

The closed hat chokes the open hat: a closed hat cuts off a ringing open hat, as on a real kit.

Voice knobs

Each row has six small knobs on a band in the voice's colour.

TUNE	Pitch, -24 to +24 semitones.
DECAY	Length, 25 - 400% of the voice's natural decay.
TONE	Brightness, 0 - 100%. What it changes depends on the model: the filter and noise colour (ANALOG), the FM index (FM), how quickly the upper modes are damped (PHYSICAL), or the phase-distortion warp and the crush rate (DIGITAL).
SNAP	The attack, 0 - 100%: the depth of the pitch sweep and the click, and in PHYSICAL how hard the mallet is.
LEVEL	The voice's volume, 0 - 100%.
PAN	Position, -100 (left) to +100 (right).

PATTERN

1 - 8	The eight pattern slots, saved with the project and in REAPER presets. Pick one to load it. While the sequencer is playing the change waits for the next bar (the slot blinks until then).
SAVE	Stores your edits into the current slot; its light shows there are unsaved edits. Right-click SAVE to store them into a different slot.
COPY / PASTE	Copies the pattern (steps and row lengths) and pastes it back, for example into another slot before saving.
CLEAR	Clears every step (the row lengths stay).
RANDOM	A musical random pattern: each voice favours the places it usually plays (kick on the beats, snare and clap on the backbeat, and so on), with some accents.
FACTORY	Loads one of the built-in patterns: House, Techno, Breakbeat, Electro, Hip-hop, Trap (32 steps), Polyrhythm or Empty.
LOAD	Loads a pattern from a text file in REAPER/Data/gfx-drum-patterns (see below).

The eight slots start out holding the eight factory patterns.

TRANSPORT

PLAY	The sequencer on or off. When on, it plays whenever REAPER's transport is playing, locked to the project tempo.
MIDI OUT	Sends the sequence out as MIDI notes, so you can record it to a MIDI item on another track.
IN	Lights when a MIDI drum note arrives.
SWING	Delays the off-beat sixteenths, 0 - 75%.
CH	The MIDI channel. OMNI (0): it listens to every channel and sends on channel 10. 1 - 16: it listens and sends on that

channel only.

COMP

A stereo-linked bus compressor on the whole kit, with a 6 dB soft knee.

THRESH	-40 to 0 dB.
RATIO	1:1 to 20:1.
ATTACK	0.1 - 100 ms.
RELEASE	10 - 1000 ms.
MAKEUP	0 to +24 dB of make-up gain.
MIX	Blends the compressed kit with the dry one, 0 - 100%. It starts at 0%, so the compressor is out until you turn MIX up; lower settings give parallel compression.
GR	The gain reduction, as a bar and a value.

MASTER

VOLUME	Output level, -24 to +12 dB.
OUT meters	Output level, left and right, with a floating peak. Click a meter to clear the peak.

Playing it from MIDI

G Pulse plays from General MIDI drum notes, with velocity. Other MIDI passes straight through.

KICK	Notes 35, 36
SNARE	Notes 38, 40
CLAP	Note 39
CL HAT	Notes 42, 44

OP HAT	Note 46
TOM	Notes 41, 43, 45, 47, 48, 50
PERC	Notes 37, 56, 75, 76, 77
CYMBAL	Notes 49, 51, 52, 55, 57, 59

MIDI OUT sends kick 36, snare 38, clap 39, closed hat 42, open hat 46, tom 45, perc 56 and cymbal 49, at velocity 127 for accents and 100 for hits.

Recording the pattern as MIDI

1. Switch MIDI OUT on.
2. Right-click the G Pulse track's record-arm button and choose Record: output > Record: output (MIDI).
3. Arm the track and record while REAPER plays. The sequence is written to a MIDI item on the track.
4. Edit the notes as you like, then switch PLAY off and set the track back to normal recording: G Pulse now plays the MIDI item.

Pattern files

LOAD reads plain text files from the folder REAPER/Data/gfx-drum-patterns (three examples are included). One line per voice: a voice name, then the steps.

- X = accent, x (or o) = hit, . (or - or _) = rest. Spaces and | between steps are ignored.
- Up to 32 steps; the row's length is the number of steps you write, so short rows make polyrhythms.
- Voice names: KICK (or BD), SNARE, CLAP (or CP), CH (closed hat, or HH), OH (open hat), TOM, PERC (or RIM, CB, COWBELL), CYMBAL (or CRASH, RIDE). A line of steps with no name goes to the next row.
- Lines starting with # are comments.
- A loaded file replaces the steps of the current pattern; press SAVE to keep it in a slot. If the file can't be opened, the PATTERN section says so.

Tips

- Try the same pattern through all four CHARACTER models - each one gives a completely different kit.
- Set a hat row to 12 steps and a perc row to 7 against a 16-step kick for patterns that keep evolving.
- Turn the bus COMP's MIX up to 30 - 50% for parallel punch without flattening the accents.
- Use accents (Shift-click) on the hats for a more human groove, then add a little SWING.

File: gfx-g-pulse.jsfx